

8-Point FFT Processor

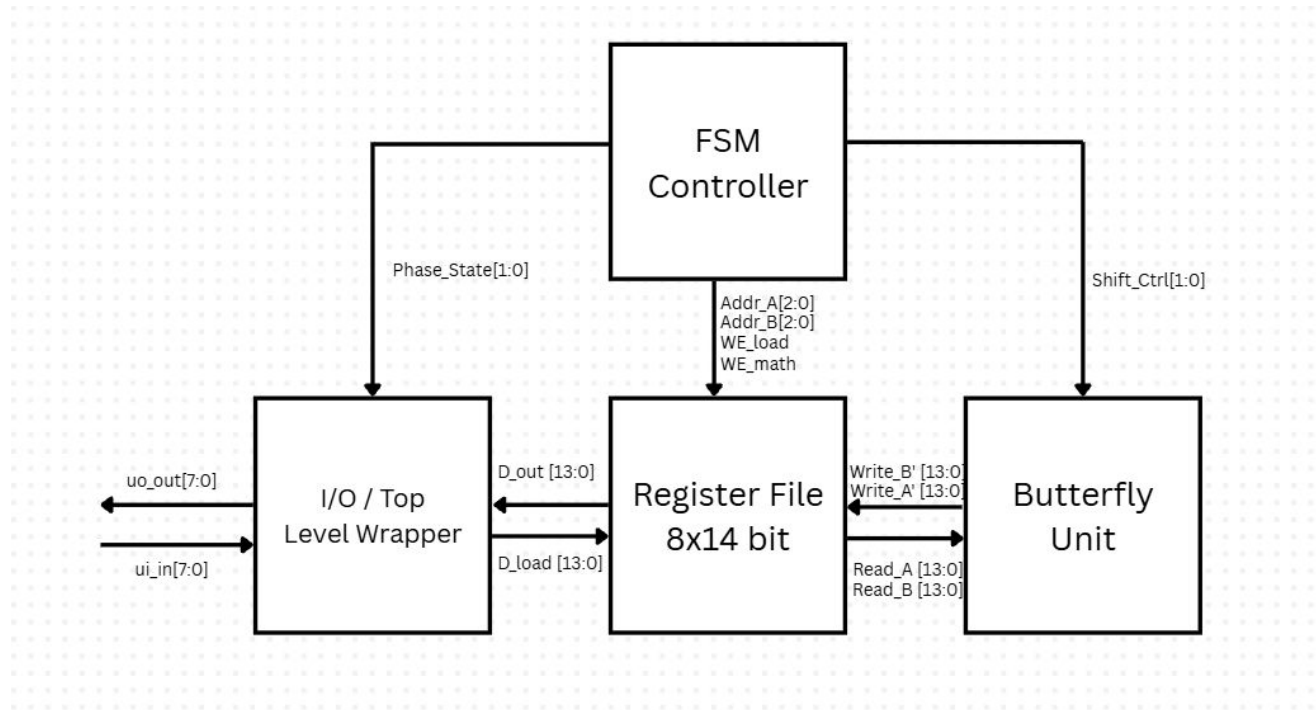
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General Overview

Fast Fourier Transform: An algorithm that converts a signal from the time domain into the frequency domain

What does 8-Point Mean? Takes 8 discrete real-valued time-domain samples and produces 8 complex-valued frequency bins

System Diagram



Specifications

Architecture Type: Radix-2 Decimation-In-Time (DIT), Folded Serial Datapath

Internal Word Size: 14 bits total per sample.

Real Component: 7 bits (1 sign bit, 6 data bits, two's complement).

Imaginary Component: 7 bits (1 sign bit, 6 data bits, two's complement).

Input Format: 8-bit time-domain samples on the `ui_in[7:0]` bus